

NurseCalc — MERN MVP Roadmap

A proper 7-day MVP plan for an educational nursing-calculation platform

Technical stack: HTML5, CSS3, JavaScript, React, Node.js, Express.js, MongoDB, Mongoose, JWT and bcrypt.

Product boundary: The MVP is an educational learning/practice product using fictional scenarios. It should not determine or authorize medication administration for real patients.

1. MVP Goal

Learn → Practice → Calculate → Understand the result → Review mistakes → Improve.

The MVP should feel complete without trying to become a full nursing platform. Prioritize calculation correctness, useful practice, and a clean student experience.

2. Core MVP Modules

Module	MVP scope
Learn	5 topics, 15 lessons, worked educational examples
Practice	Topic/difficulty selection, questions, answers, explanations
Calculator	Validated educational calculation engine with unit handling
Mistakes	Classify errors such as conversion, arithmetic, formula and rounding
Progress	Accuracy, attempts, topic performance, weak-area recommendations
AI	One feature: Explain My Mistake; AI does not perform the calculation

3. Seven-Day Development Plan

Day 1 — Foundation

Set up React frontend, Express backend, MongoDB/Mongoose, environment variables, authentication models, API structure and responsive navigation.

Day 2 — Learn

Create 5 topics: Medication Math Basics, Unit Conversions, Tablet Calculations, Liquid Calculations and IV Flow Mathematics. Build lessons, completion state and topic pages.

Day 3 — Calculation Engine

Build input validation, unit normalization, deterministic formulas, rounding rules and independent verification. Write automated JavaScript tests before expanding features.

Day 4 — Practice

Build question selection, answer submission, result screen, worked explanation and attempt storage. Start with 100 reviewed educational questions.

Day 5 — Mistakes + Progress

Store attempts, classify mistakes, calculate topic accuracy and show weak areas with targeted practice.

Day 6 — AI + Polish

Add Explain My Mistake, mobile polish, loading/error/empty states, accessibility, API validation and safety messaging.

Day 7 — QA + Beta

Run the complete user flow, test calculation edge cases, test desktop/mobile, fix critical bugs and prepare a small student beta.

4. Recommended Project Structure

```
nursecalc/  
  ■■■ client/ — React, HTML, CSS, JavaScript  
  ■■■ server/  
    ■ ■■■ server.js  
    ■ ■■■ config/  
    ■ ■■■ models/  
    ■ ■■■ routes/  
    ■ ■■■ controllers/  
    ■ ■■■ middleware/  
    ■ ■■■ services/  
  ■■■ package.json  
  ■■■ .env
```

5. MongoDB Collections

users, **topics**, **lessons**, **questions**, **attempts**, **mistakes**, **progress** and **bookmarks**. Keep the initial schema small; add teacher, class, assignment and subscription collections later.

6. Accuracy Architecture

```
User input
↓
Validation
↓
Unit normalization
↓
Deterministic calculation engine
↓
Independent verification
↓
Rounding rule
↓
Educational result
↓
AI explanation (optional)
```

AI should explain an already-verified result rather than decide the mathematical result. Every supported calculation should have automated tests covering normal values, decimals, unit conversions, invalid input, edge cases and rounding.

7. Content/Data Pipeline

```
Authoritative educational references → original lesson/question draft → deterministic
math validation → nursing/clinical review → approval → MongoDB → student practice.
```

Use recognized nursing education resources and official health/medication references for research. Create original explanations and questions rather than copying competitor content. AI-generated questions must remain drafts until validated and reviewed.

8. MVP Acceptance Checklist

- Registration/login works
- 5 topics and 15 lessons are usable
- Calculator validates inputs and units
- Calculation engine has automated tests
- 100+ questions are reviewed
- Practice results and explanations work
- Wrong answers create mistake records
- Progress updates correctly
- AI only explains approved/verified educational results
- Responsive mobile UI works
- Environment secrets are not exposed
- API input validation and authentication are enabled
- Clear educational-use disclaimer is visible
- Privacy policy and terms are prepared before public launch

9. After MVP

V1.1: adaptive learning → V1.2: mock exams → V1.3: gamification → V1.4: teacher dashboard → V1.5: college accounts → V2: subscriptions and advanced AI.

The first success metric should be repeat learning, not raw downloads: for example, recruit 50–100 nursing students, measure second-session retention, questions completed, accuracy improvement and weak-topic recovery.

Launch principle: Build fewer features, but make the calculation engine, content quality, student workflow and safety boundary excellent.